OpenCL/OpenGL Interoperation + MeshFree FEM

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  - Computations like this tend to be hacks

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- Often better-suited for vectorized CPU code (SSE, AVX) than C and/or intrinsics

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  - Many more: see your local HPC class

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- Transferring data to/from device adds significant overhead!
- Note: when the device is a CPU, all copies can be eliminated...

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- This memory could be VBOs (vertex buffers), textures on the OpenGL side.
- We'll show how to have OpenCL write to a texture that OpenGL draws to the screen.

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```
CGLContextObj kCGLContext = CGLGetCurrentContext();
CGLShareGroupObj kCGLShareGroup = CGLGetShareGroup(kCGLContext);

cl_context_properties props[] = {
    CL_CONTEXT_PROPERTY_USE_CGL_SHAREGROUP_APPLE,
    (cl_context_properties) kCGLShareGroup,
    0
};

cl_int status;
context = clCreateContext(props, 0, 0, NULL, 0, &status);
```

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```
GLuint tex;
glGenTextures(1, &tex);
glBindTexture(GL_TEXTURE_2D, tex);
glTexImage2D(GL_TEXTURE_2D, 0, GL_RGBA, width, height, 0, GL_RGBA,
             GL_UNSIGNED_BYTE, NULL);
d_img = clCreateFromGLTexture(context, CL_MEM_READ_WRITE, GL_TEXTURE_2D,
        0, tex, &err);
```

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```
glFinish();
clEnqueueAcquireGLObjects(queue, 1, &d_img, 0, NULL, NULL);
clEnqueueNDRangeKernel(queue, kernel, 2, NULL, gdim, ldim,
                       0, NULL, NULL));
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- glFinish/clFinish ensure GL/CL commands each finish before the other is run
- Benchmark warning: without glFinish, clEnqueueAcquireGLObjects takes non-negligible time

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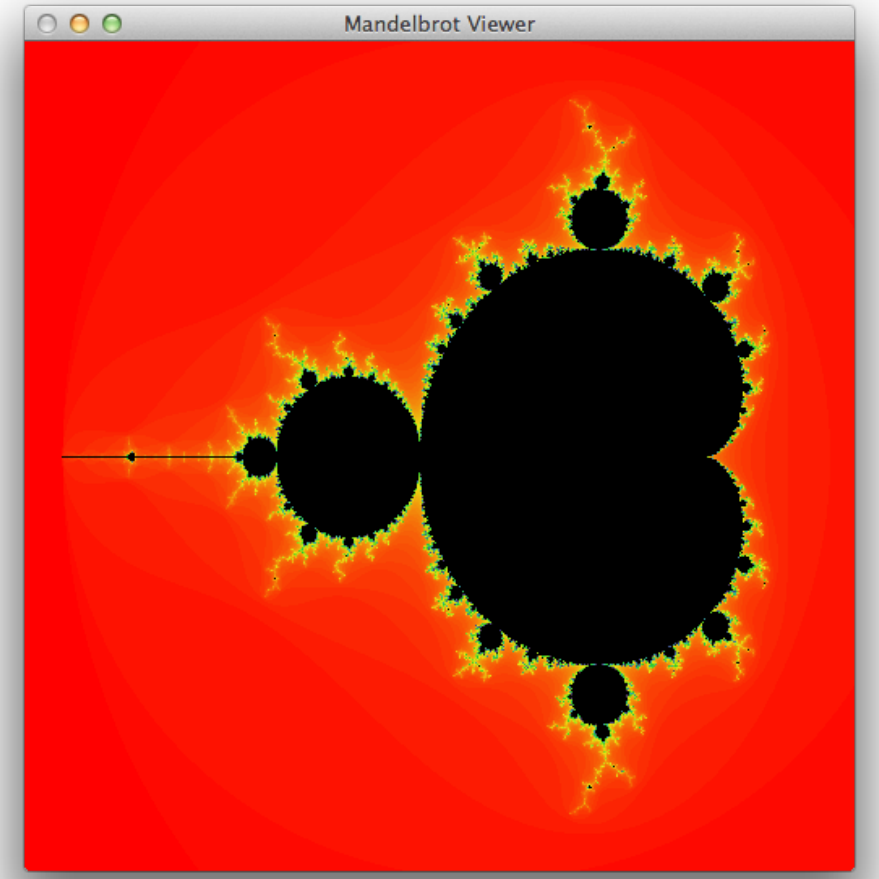
- The OpenCL buffer bound to the texture is of type `image2d_t`
- must use `write_imagef` to write to the texture.

```
__kernel void mandelbrot(write_only image2d_t img, int w, int h,
                         float xMin, float xMax, float yMin, float yMax,
                         __global float *colorMap, unsigned int maxIters)
{
    ...
        write_imagef(img, (int2)(col, row),
                     (float4)(colorMap[colorOffset + 0],
                              colorMap[colorOffset + 1],
                              colorMap[colorOffset + 2], 1.0f));
    ...
}
```

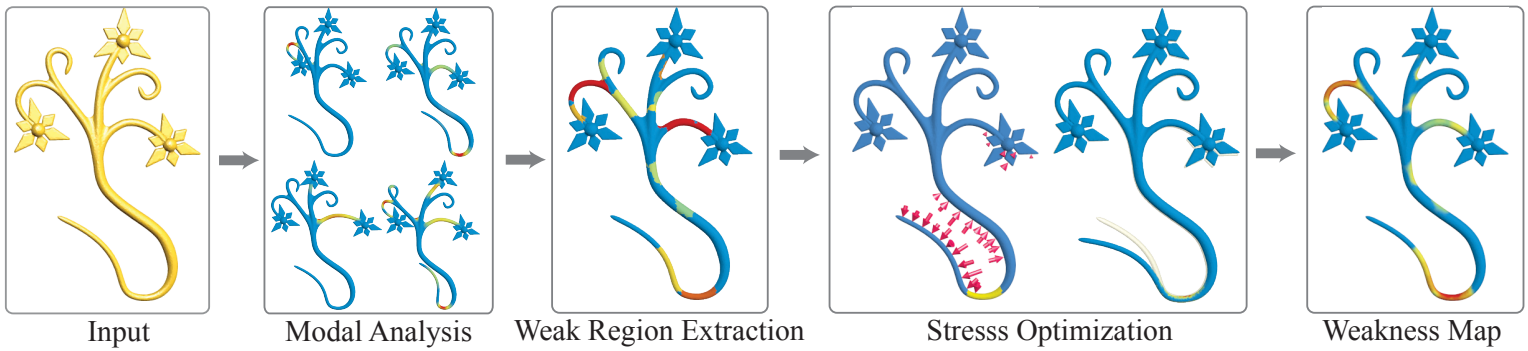
Demo: OpenCL Mandelbrot Viewer



Demo

Mesh-free FEM

- James recently described our work on structural analysis.
- Pipeline:



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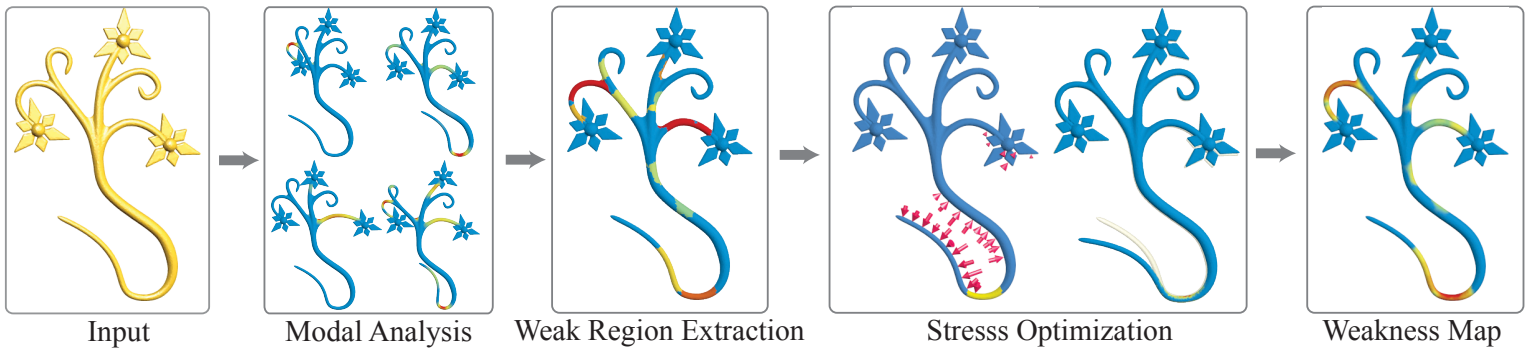

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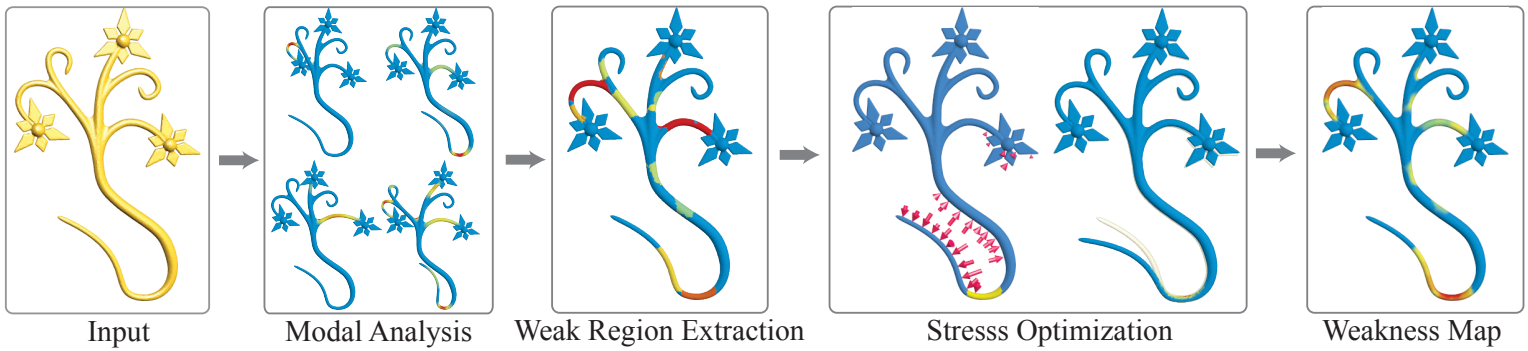
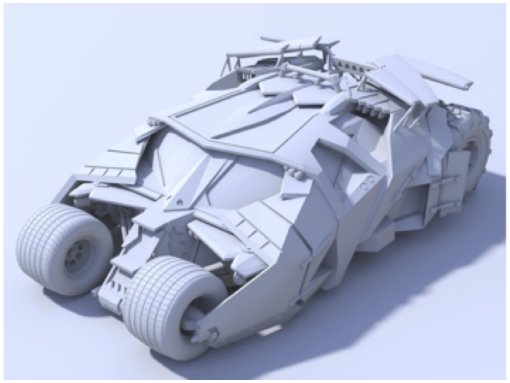


- Missing initial stage: volume meshing!
- There are several reasons to want to avoid meshing.

Mesh-free FEM: Why?

- Complicated geometry can be hard to mesh robustly.



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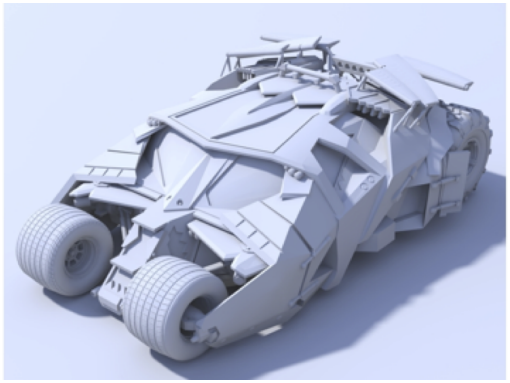

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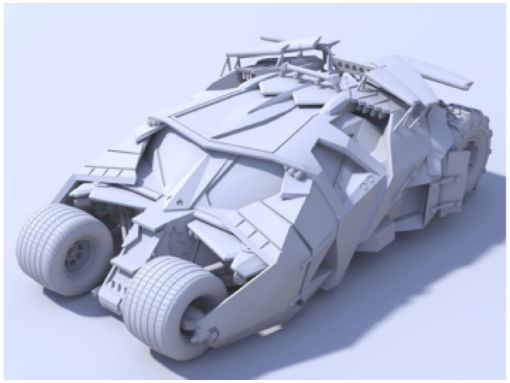

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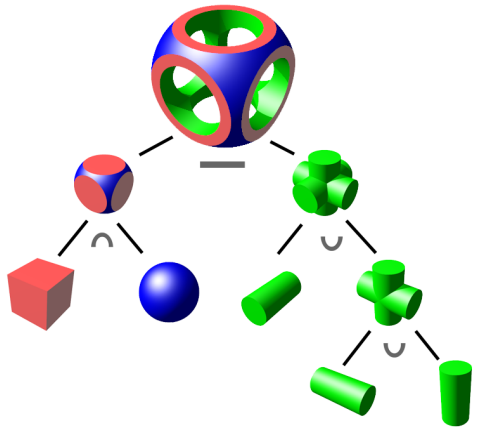
- Sometimes you don't even have a mesh (CSG)



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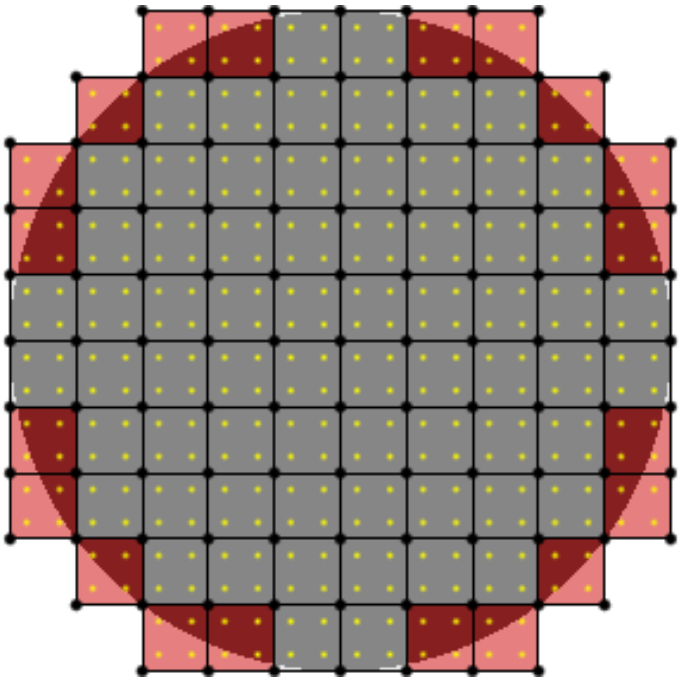

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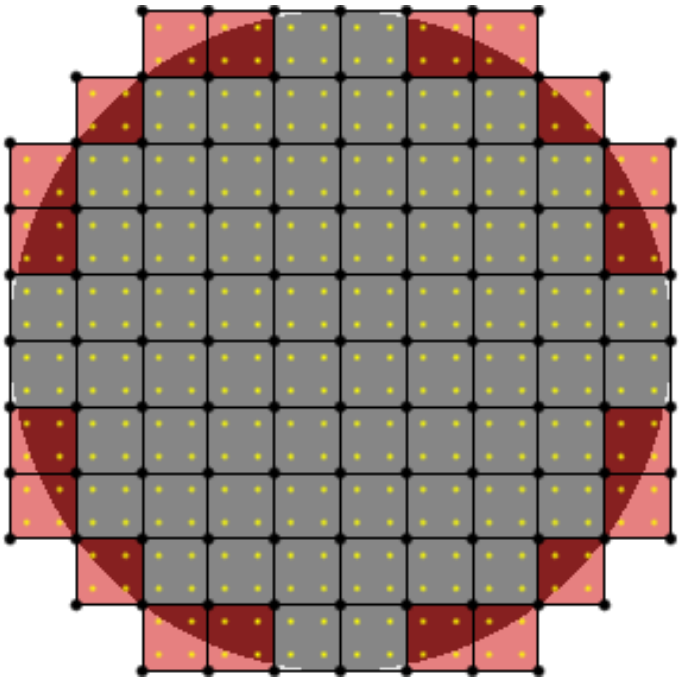

- Note: doesn't conform to the object's boundary.

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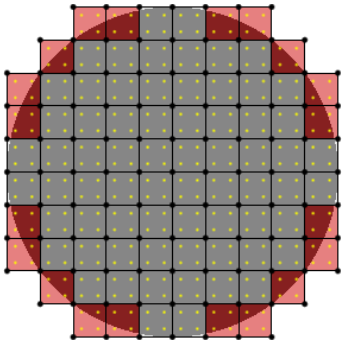

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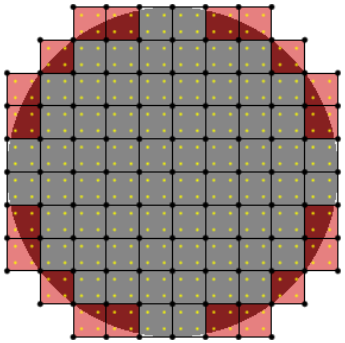

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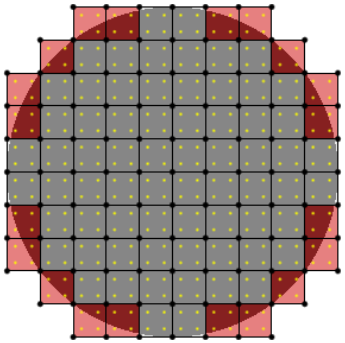

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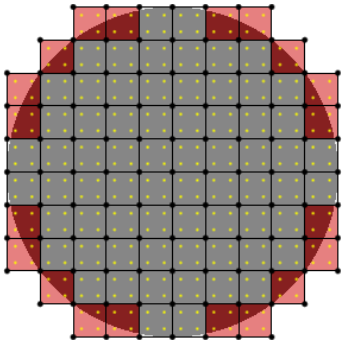

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- Integrands are not continuous on boundary cells (defined to be zero outside object) $\Longrightarrow$ many quadrature points needed

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  - We plan to blur the surface forces to convert them into volume forces that can be integrated over the cells

Implementation: CSGFEM

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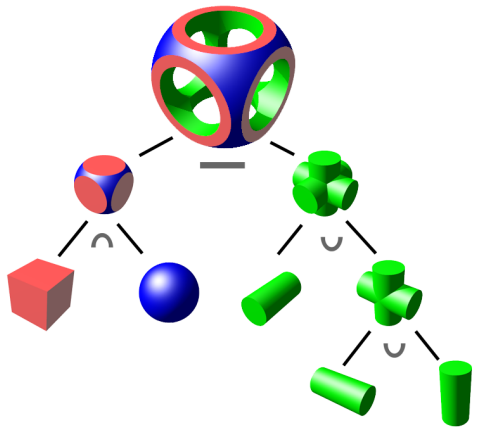

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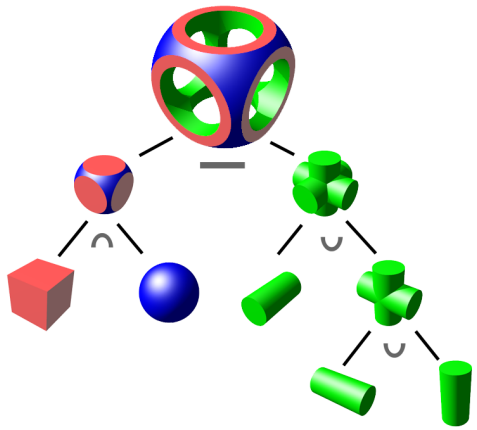

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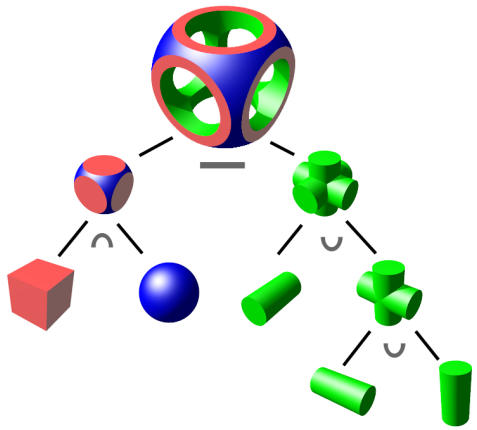

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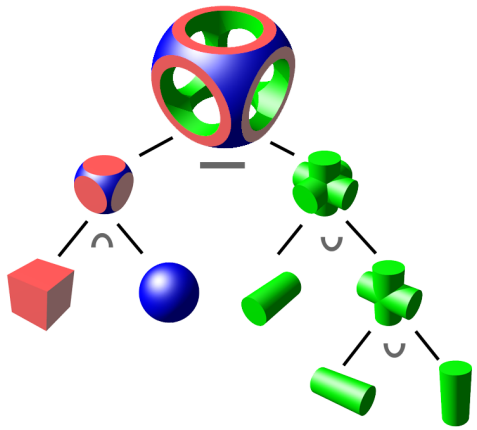

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- Currently have modal analysis working...

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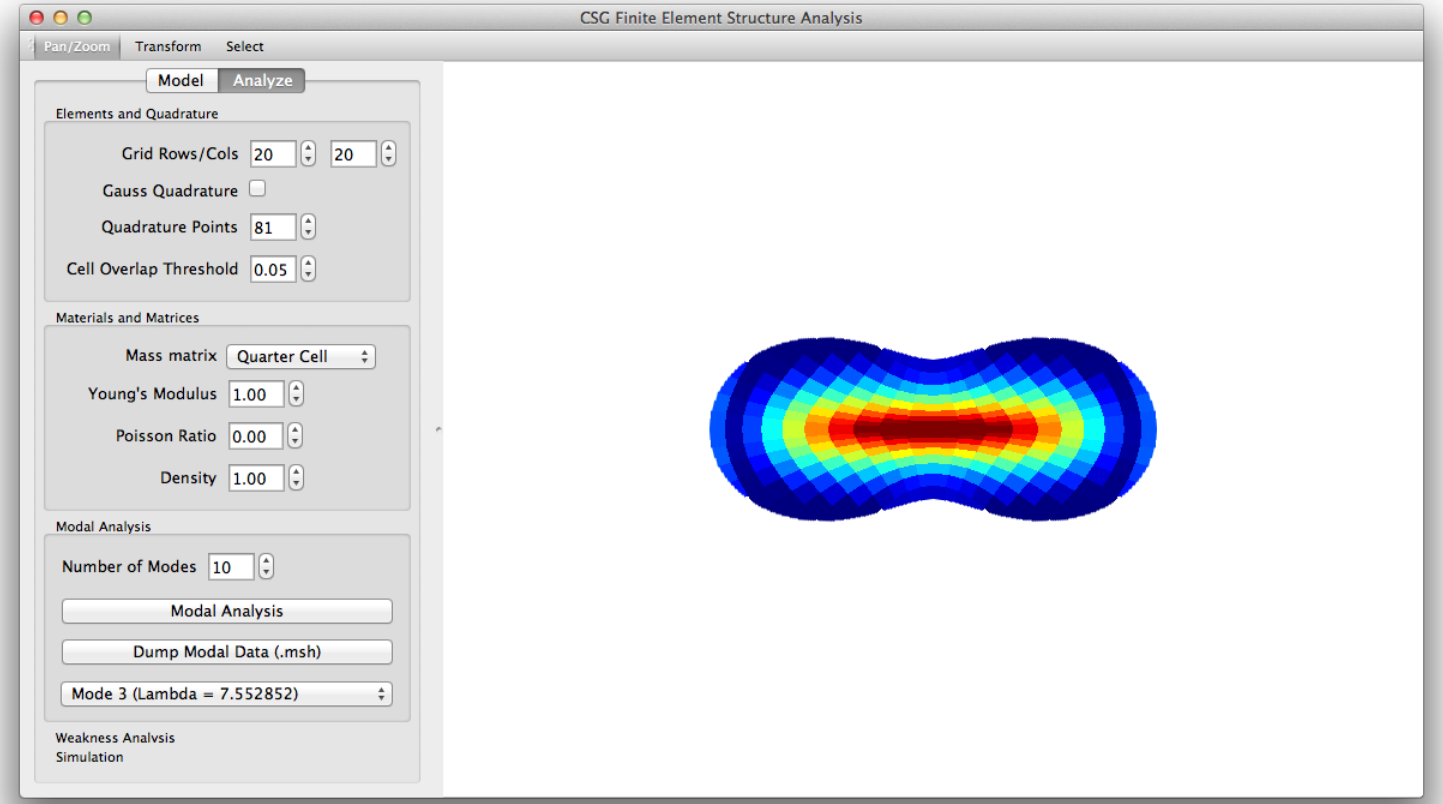

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